



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

SECJ 2203: Software Engineering

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USE CASE

COURSE REGISTRATION SYSTEM

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1. Use Case Diagram

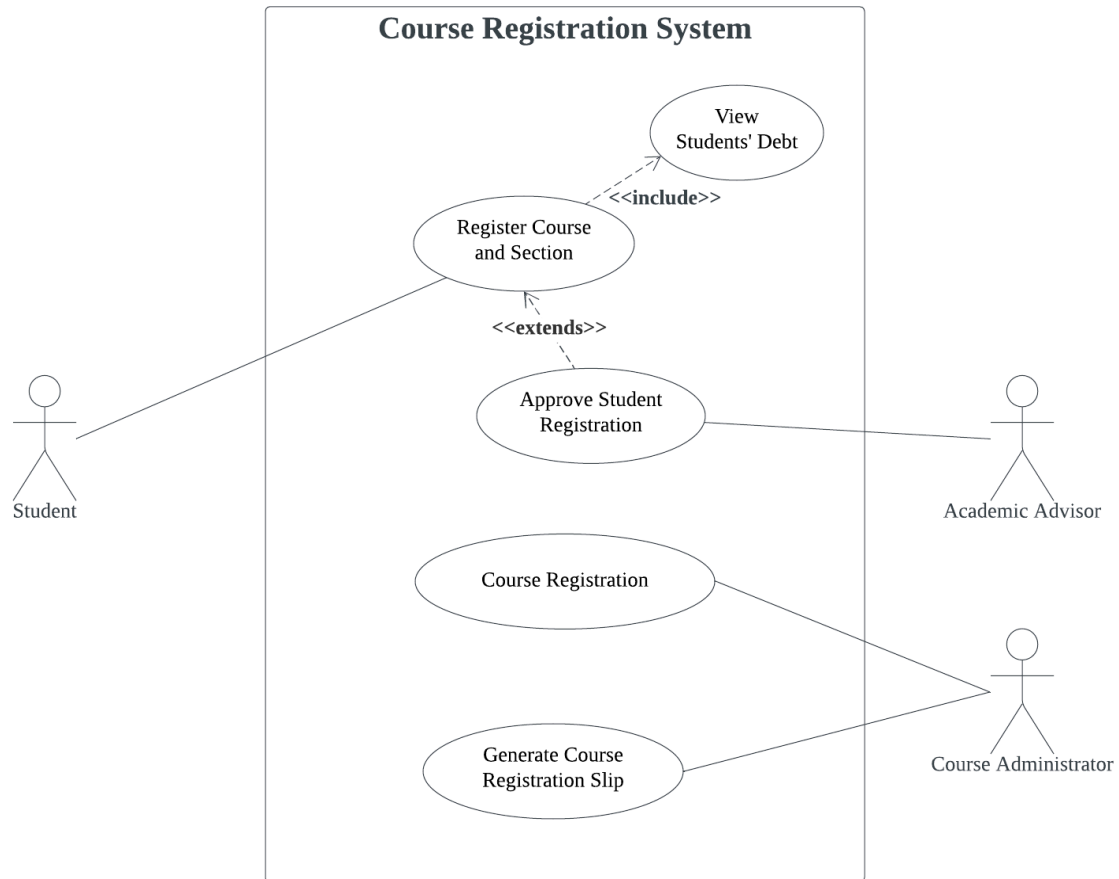


Figure 1.1 Use Case Diagram

2. Class Diagram

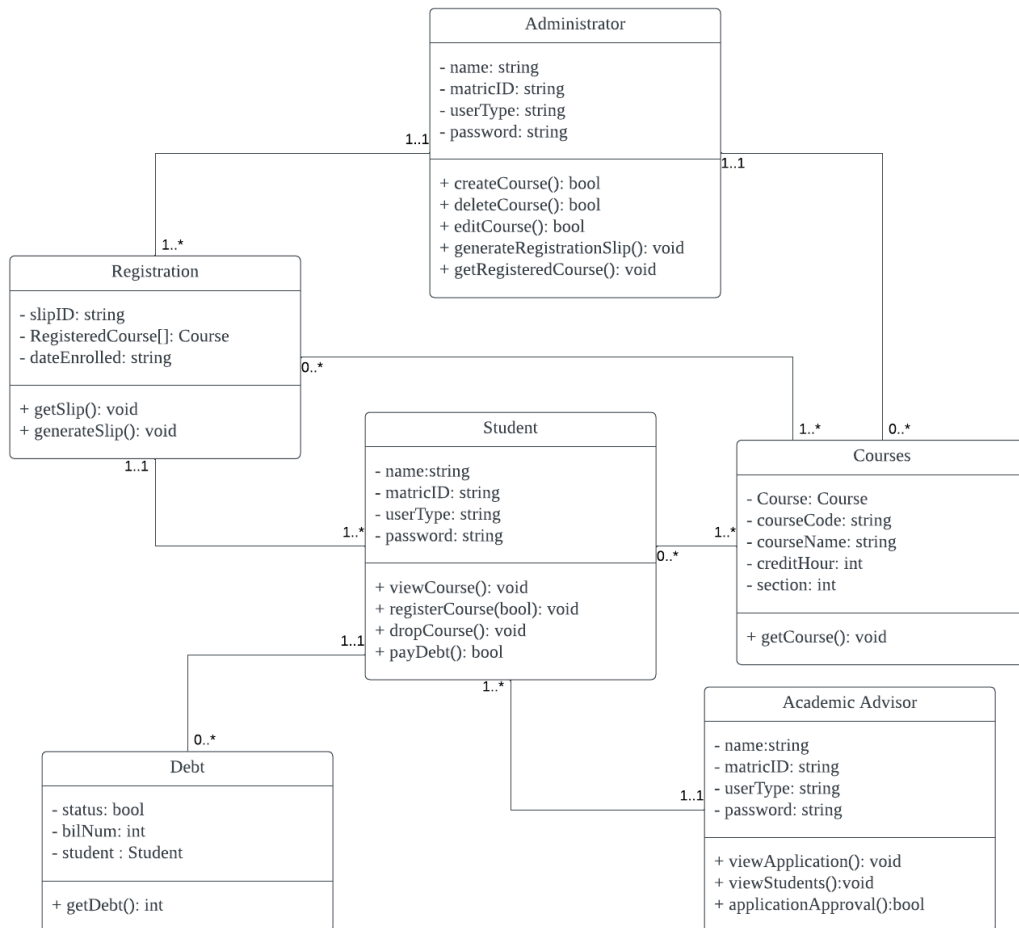


Figure 1.2 Class Diagram

Use Case Description, Activity Diagram & Sequence Diagram

- Use Case 1

Use case: View Students' Debt
ID: UC2
Actors: -
Brief Description: This use case describes how the student can view their own debt through the system.
Preconditions: Users have unpaid fees.
Flow of events: 1. The system displays the amount of student debt interface.
Postconditions: 1. Students can pay tuition fees to clear the debt.
Alternative flow 1: 1. At any time students can exit the course registration process.
Postconditions:

Figure 1.3.1 Use Case 2 : Description

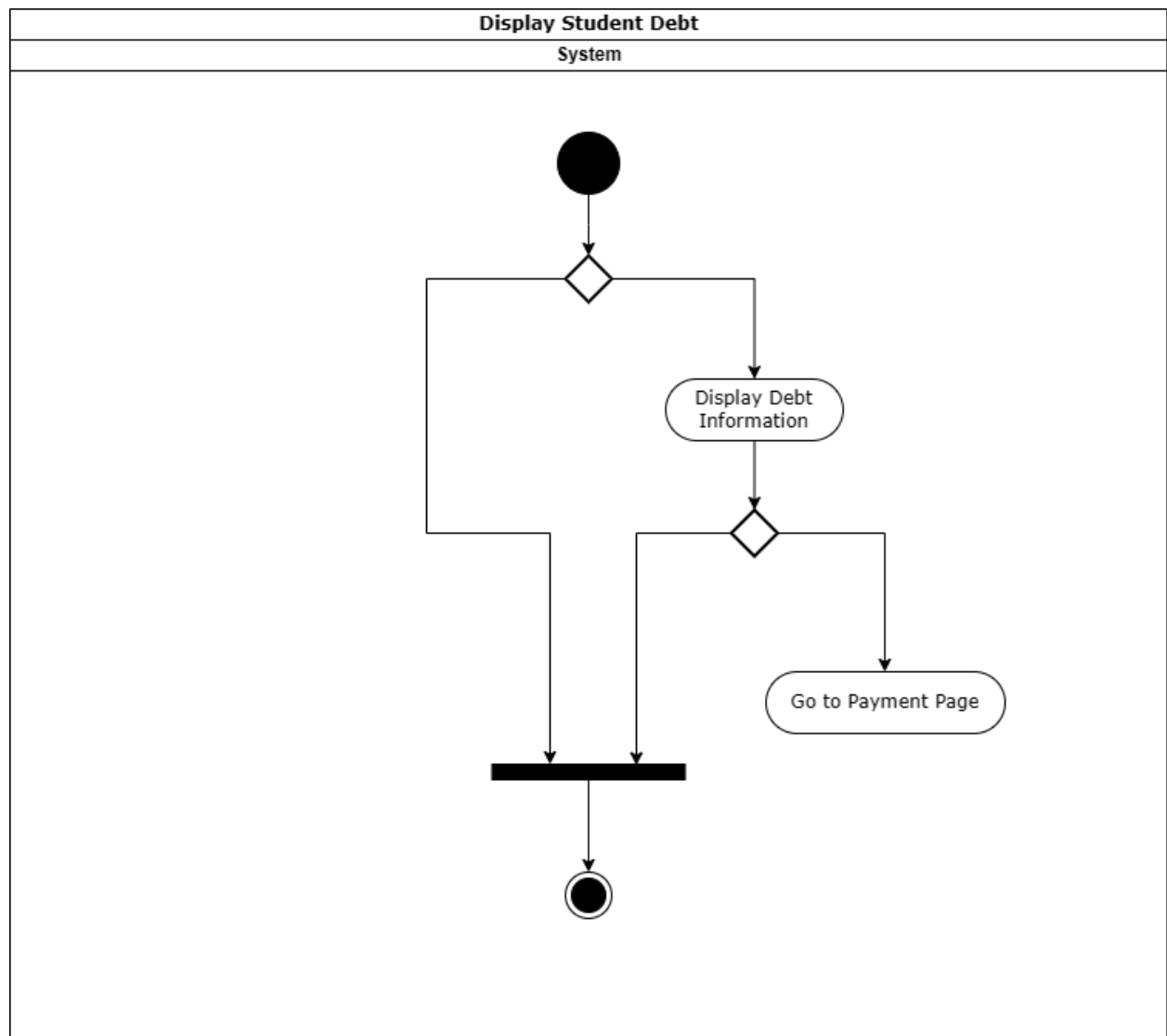


Figure 1.3.2 Use Case 2 : Activity Diagram

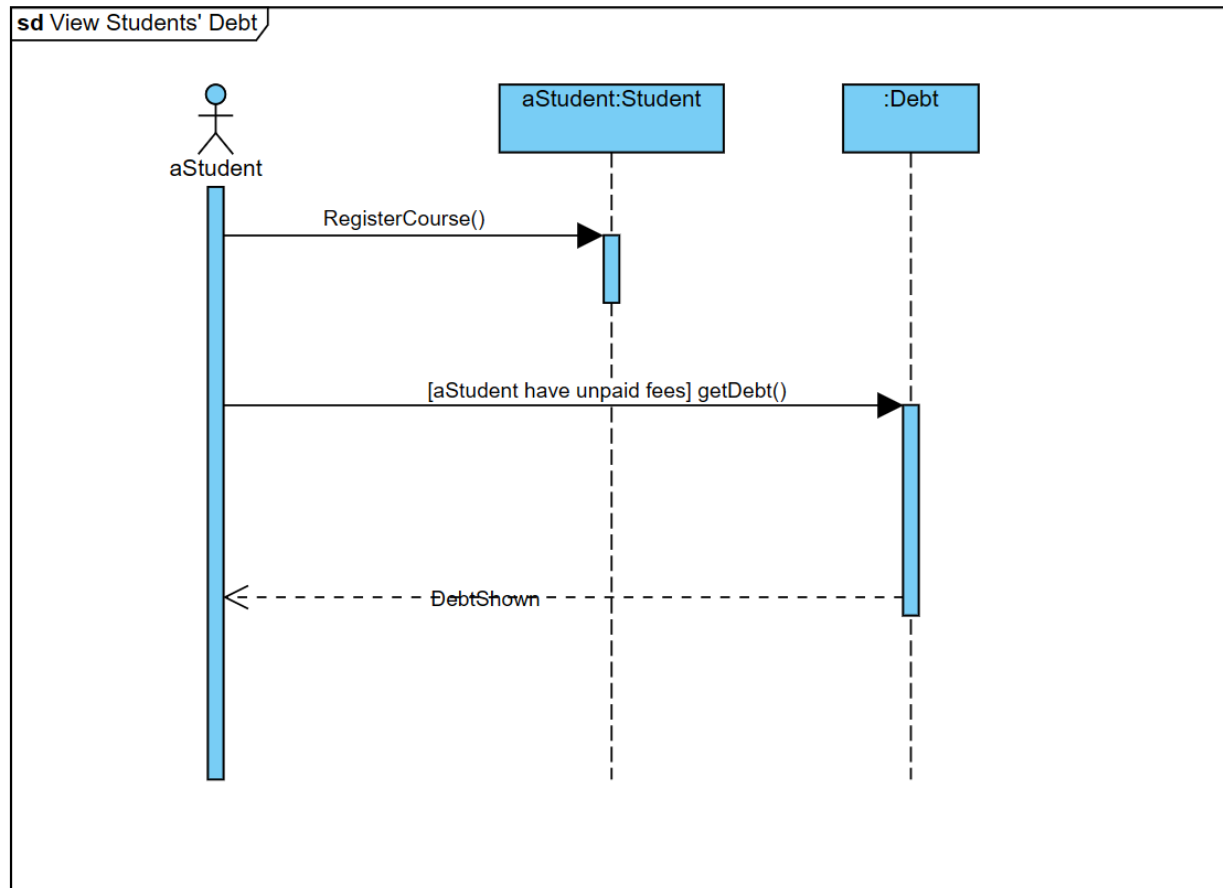


Figure 1.3.3 Use Case 2 : Sequence Diagram

- **Use Case 2**

Use case: Register Course & Section
ID: UC1
Actors: Student
Preconditions: 1. Internet access is available. 2. User login as UTM student.
Flow of events: 1. Students select course registration from MyUTM Portal dashboard. 2. Students enter course code and section. 3. Students save and submit registered courses.
Postconditions: 1. System shows status as pending approval.
Alternative Flow 2: 1. At any time students can exit the course registration screen.
Postconditions:

Figure 1.4.1 Use Case 1 : Description

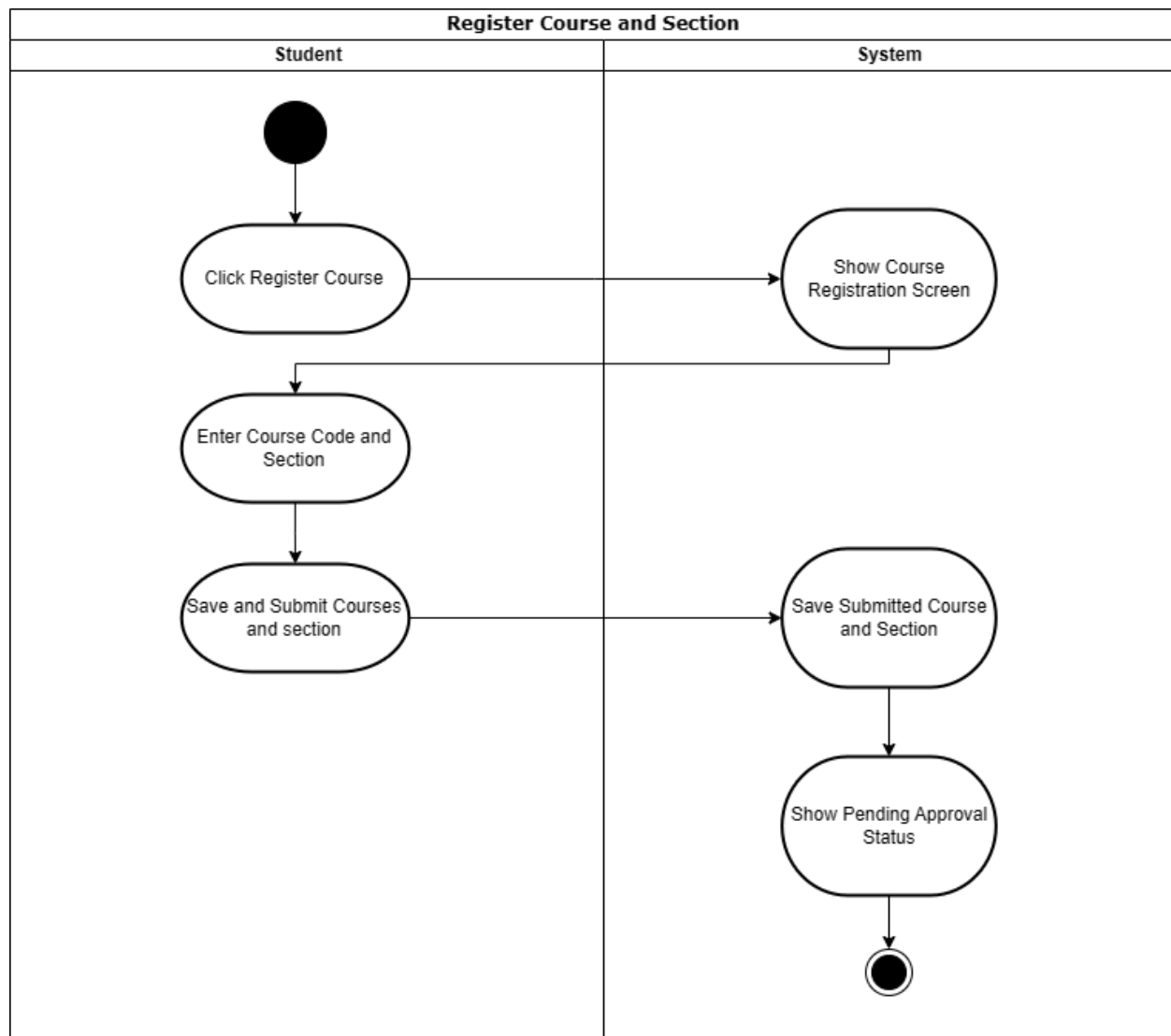


Figure 1.4.2 Use Case 3 : Activity Diagram

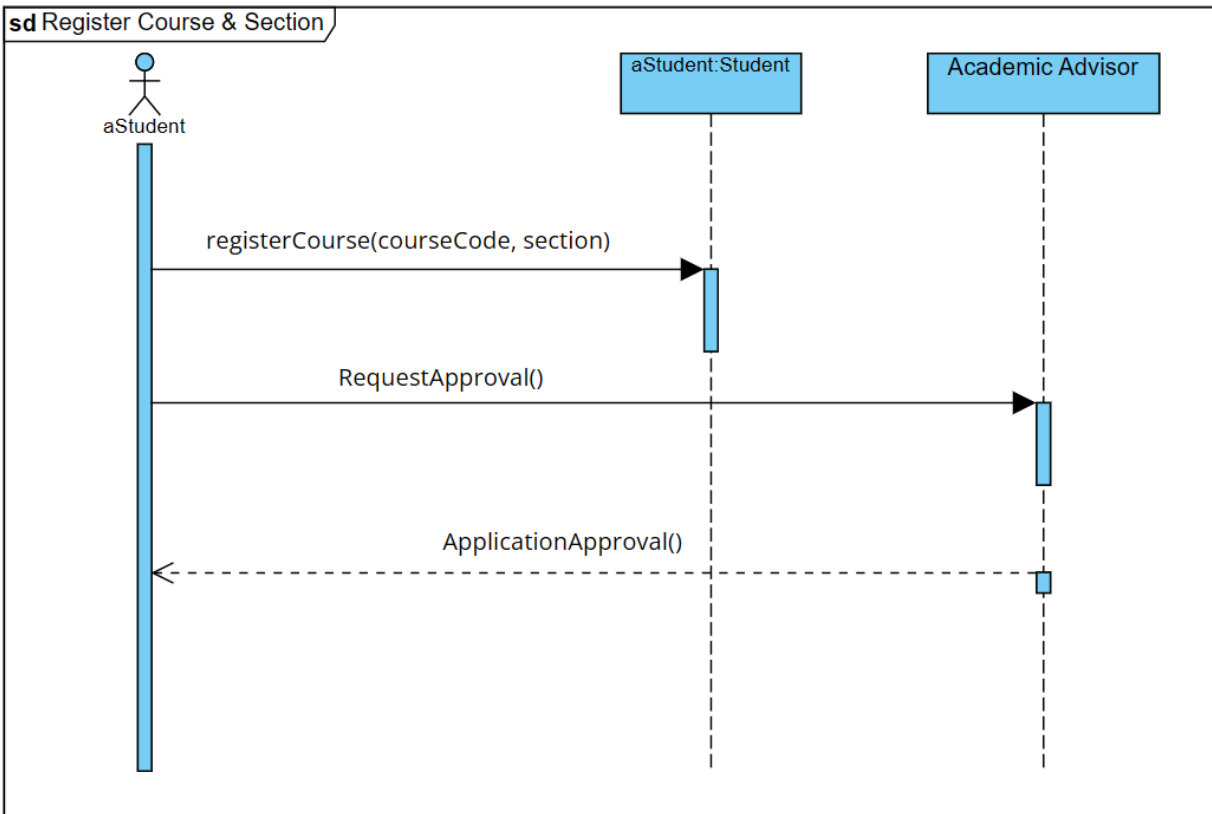


Figure 1.4.3 Use Case 1 : Sequence Diagram

- **Use Case 3**

Use case: Approve Student Registration
ID: UC3
Actors: Academic Advisor Student
Preconditions: 1. The student has completed the course registration process. 2. The system shows the student's registration details.
Flow of events: 1. The academic advisor logs into the course registration system. 2. The academic advisor views the student's registration request. 3. The academic advisor reviews the selected courses and verifies if the student meets the necessary requirements. 4. If the registration details are correct and meet all criteria, the academic advisor approves the registration. 5. The system updates the student's registration status to "Approved" and notifies the student. 6. If there are issues with the registration (e.g., missing prerequisites or course conflicts), the advisor may request the student to make changes before approval.
Postconditions: 1. The student's course registration is either approved, or feedback is provided for revisions.

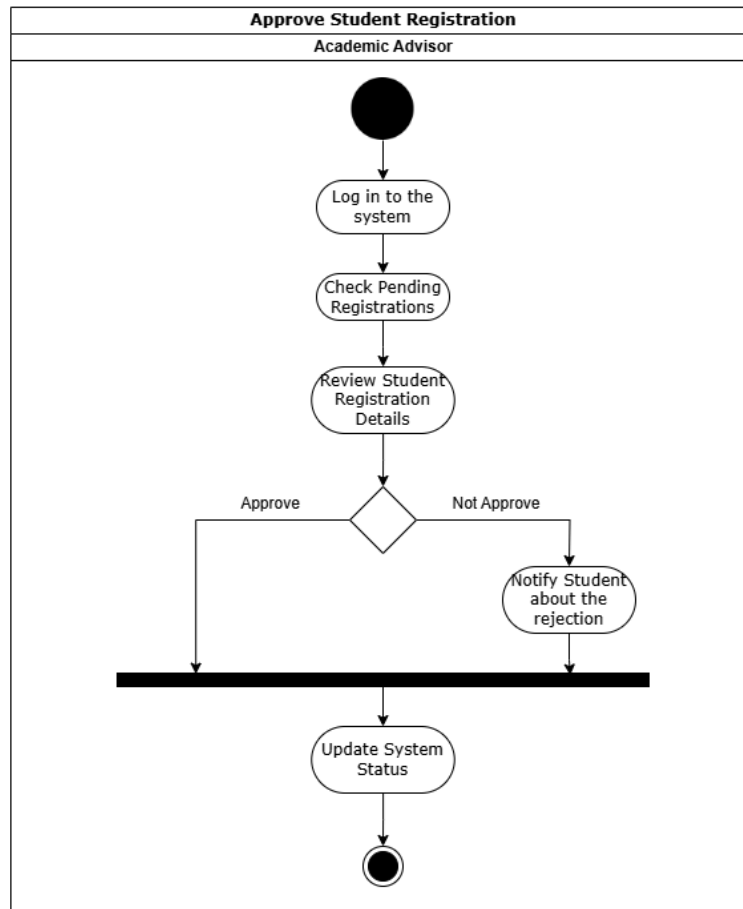


Figure 1.5.2 Use Case 3 : Activity Diagram

Approve Student Registration

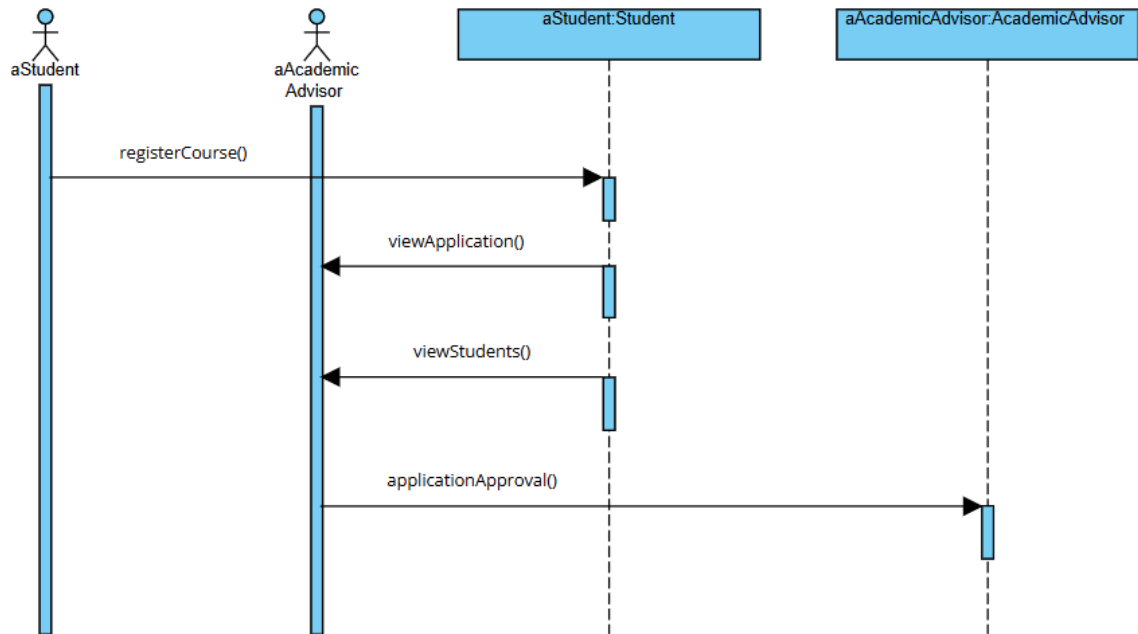


Figure 1.5.3 Use Case 3 : Sequence Diagram

- Use Case 4

Use Case : Manage Course Registration
ID : UC4
Actors : ● Course Administrator ● System ● Student
Preconditions : 1. Course quotas are set to match the assigned number of students, so no over-enrollment occurs unless the student selects an incorrect course.
Flow of events : 1. The course administrator checks for any possible errors. 2. The course administrator finalizes and verifies the approved course registration. 3. The course administrator triggers the system to notify students that their course registration is accepted.
Alternative flow : 1. If the student is repeating a semester, and their previous courses overlap with current course timings: The system informs the student to consult the Course Administrator for manual adjustment. 2. If the student accidentally registers for courses not assigned to their semester: The system alerts the student and prompts them to select and register the correct courses for their current semester.
Postconditions : 1. The system sends notifications to the student regarding their course registration status.

Figure 1.6.1 Use Case 4 : Description

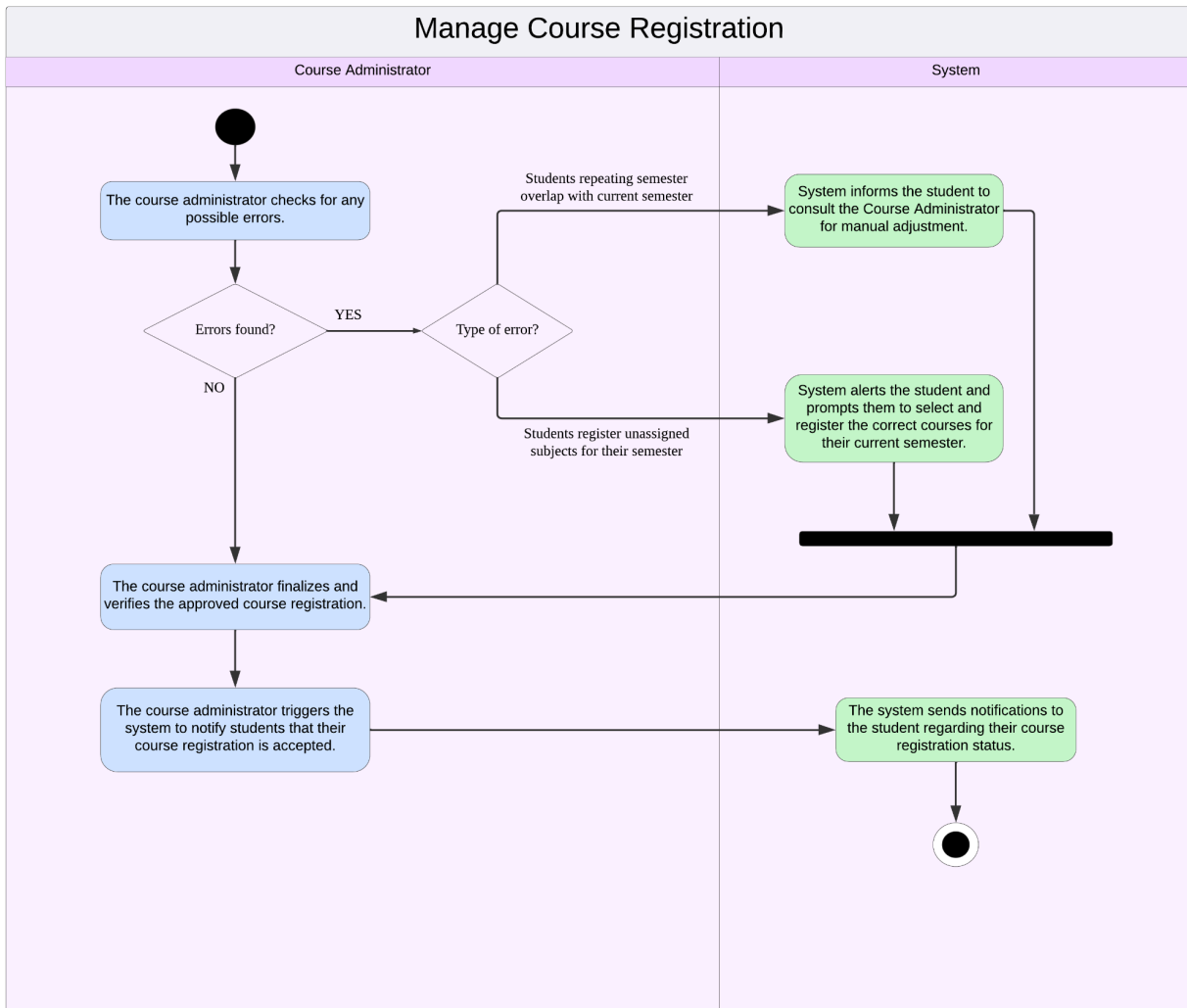


Figure 1.6.2 Use Case 4 : Activity Diagram

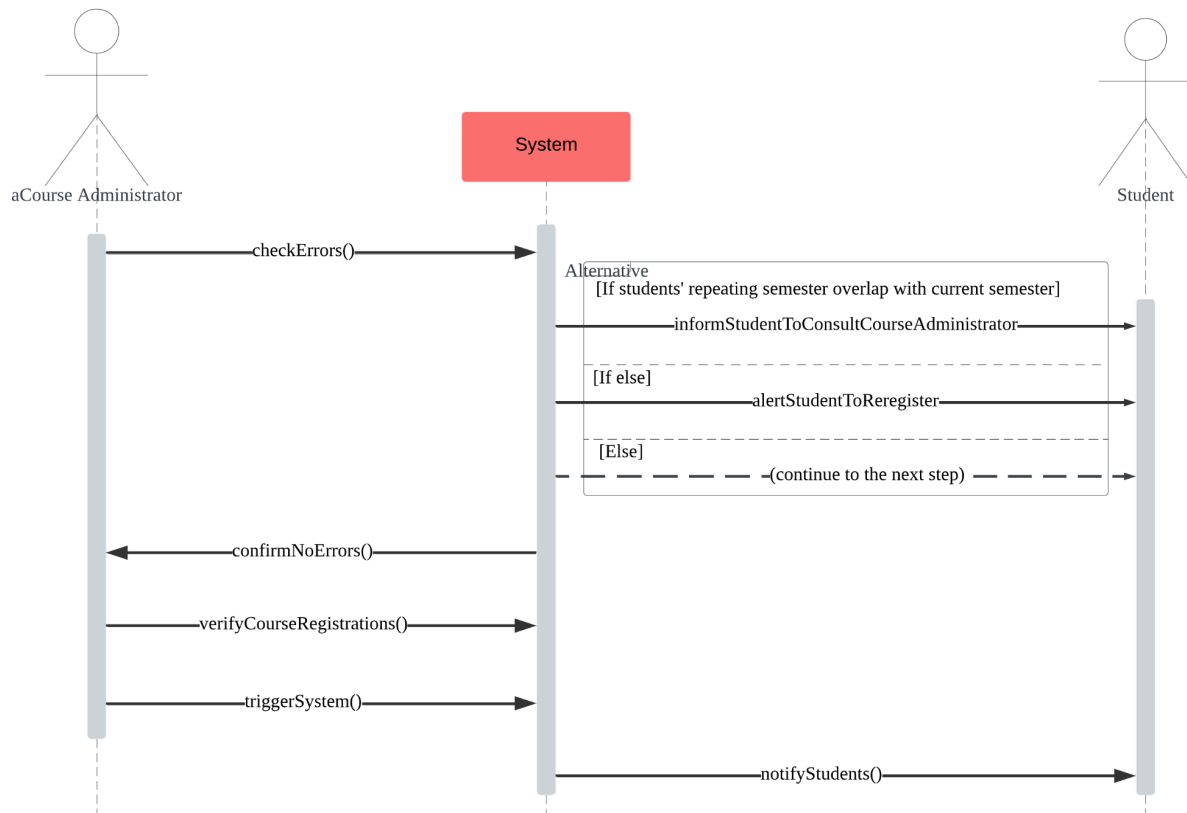


Figure 1.6.3 Use Case 4 : Sequence Diagram

- **Use Case 5**

Use Case : Generate Course Registration Slip	
ID : UC5	
Actors : ● Course Administrator ● System ● Student	
Preconditions : 1. The course administrator already triggers the system to notify students' course registration status.	
Flow of events : 1. The Course Administrator selects students with accepted course registrations. 2. The system retrieves the student's registration details, including: 4.1 Student information (name, ID number, etc.) 4.2 List of registered courses (course codes, course names, credits) 4.3 Semester and academic year 3. The Course Administrator requests the system to generate a course registration slip. 4. The system generates a digital course registration slip containing the necessary details and displays it to the Course Administrator.	
Alternative flow : 1. If there were students who didn't(completed) register: the system displays an appropriate message to the Course Administrator.	
Postconditions : 1. The Course Administrator distributes the registration slip to the student via email.	

Figure 1.7.1 Use Case 5 : Description

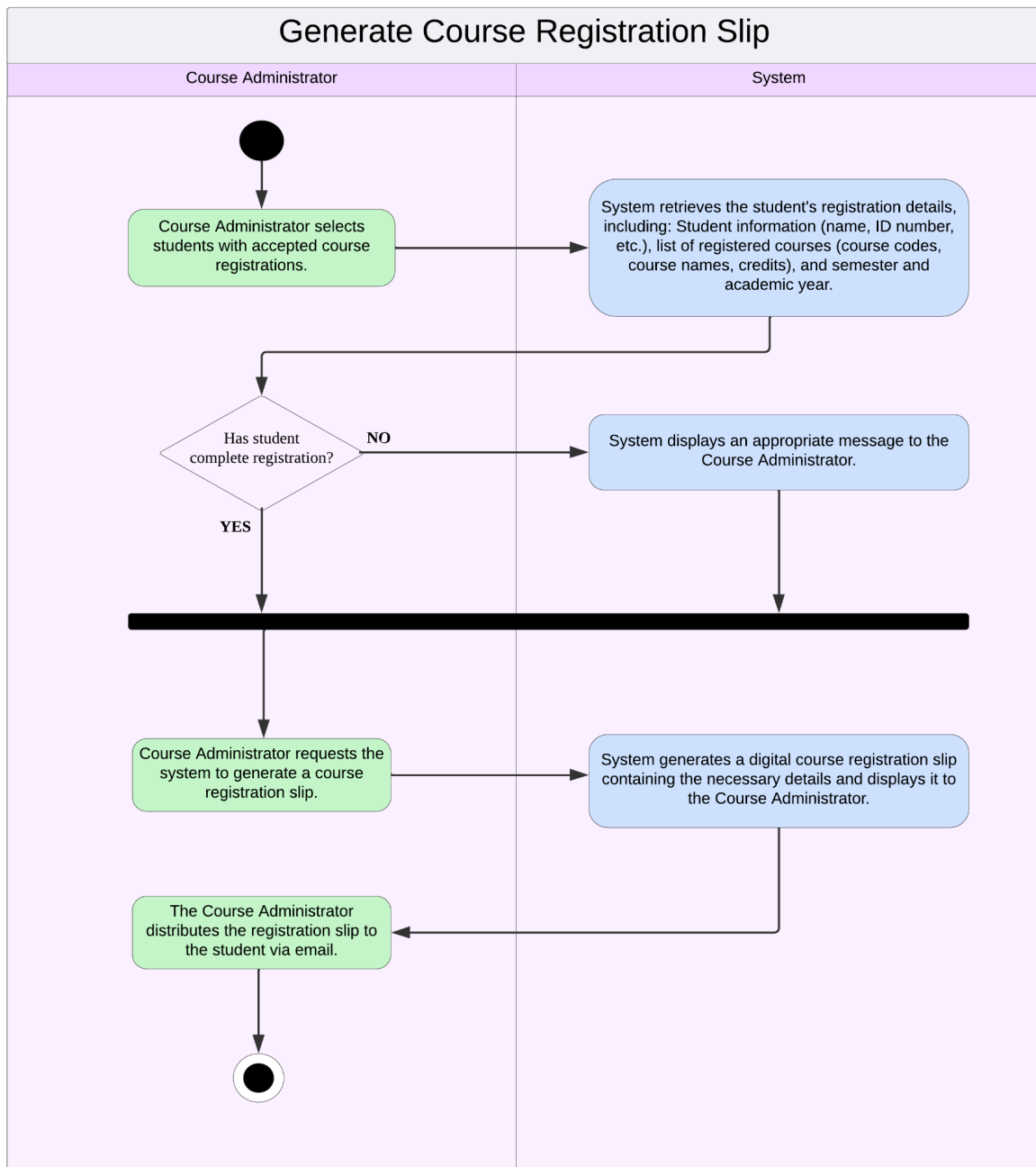


Figure 1.7.2 Use Case 4 : Activity Diagram

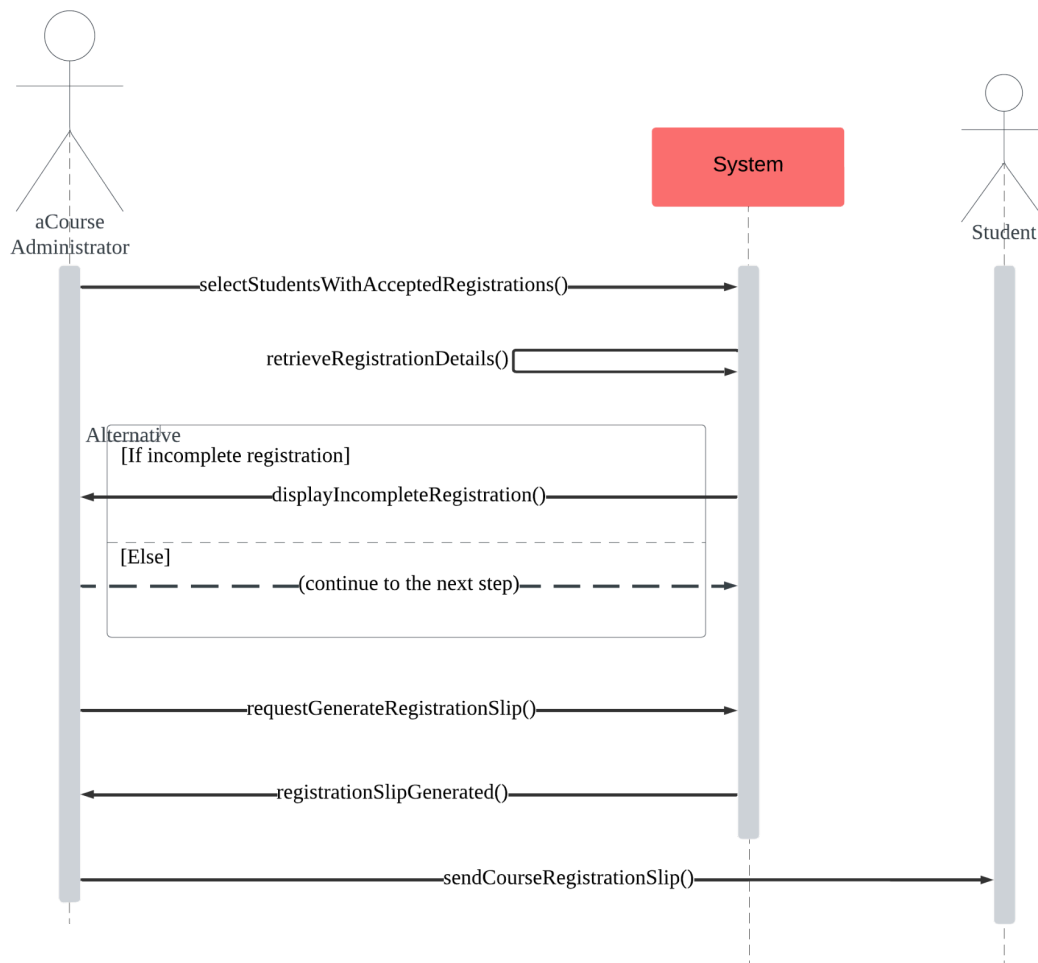


Figure 1.7.3 Use Case 5 : Sequence Diagram